

Django Tutorial #18: Bootstrap Integration

Instructor: Mohit Decodes | Comprehensive Class Notes

1. Introduction to Bootstrap in Django

Bootstrap ek popular CSS framework hai jo developers ko styling jaldi likhne aur responsive web development fast karne mein madad karta hai.

Is Tutorial mein 3 mukhy tarike (methods) sikhaye gaye hain:

- Method 1: CDN Method (Fastest & Easiest)
- Method 2: Downloaded Files Method (Offline support)
- Method 3: PIP Package Method (`django-bootstrap-5`)

2. Setting Up Environment & Project

Instructor ne ek existing Django project aur app structure ke sath shuruat ki hai:

- **Virtual Environment Create karna:** `python3 -m venv venv`
- **Virtual Environment Activate karna (Mac/Linux):** `source venv/bin/activate`
- **Django Installation:** `pip install django`

3. Method 1: Using Bootstrap via CDN

Is method mein bina koi file download kiye direct official links use kiye jate hain.

- **Step 1:** getbootstrap.com par jayein.
- **Step 2:** CSS ka `<link>` tag copy karke base HTML template ke `<head>` ke andar paste karein.
- **Step 3:** JS ka `<script>` tag copy karke `</body>` se just pehle paste karein.

Pros

- Bohot fast aur aasan setup.
- Hamesha latest version milta hai.
- Heavy files store nahi karni padti.

Cons

- Internet connection zaroori hai.
- CDN slow hone par page load time par asar pad sakta hai.

4. Method 2: Using Downloaded Bootstrap Files (Offline)

Is method mein compiled CSS aur JS files download karke project ke static folder mein rakhi jati hain.

- **Step 1:** GetBootstrap se compiled ZIP download karein.
- **Step 2:** `bootstrap.min.css` ko static/css folder mein rakhein.
- **Step 3:** `bootstrap.bundle.min.js` ko static/js folder mein rakhein.
- **Step 4:** Base template mein Django ke static tag se link karein:
`<link rel="stylesheet" href="{% static 'css/bootstrap.min.css' %}">`

Pros

- Completely offline kaam karta hai.
- Custom modifications allow karta hai.

Cons

- Updates manually download karne padte hain.

5. Method 3: Installing Bootstrap via PIP

Django ke liye specially banaye gaye python package ka use karna:

- **Step 1:** Terminal mein run karein: `pip install django-bootstrap-5`
- **Step 2:** `settings.py` mein `'django_bootstrap5'` ko `INSTALLED_APPS` mein add karein.
- **Step 3:** Base template mein load karein: `{% load django_bootstrap5 %}`
- **Step 4:** Template tags use karein: `{% bootstrap_css %}` aur `{% bootstrap_javascript %}`

6. Live Testing & Summary

Teeno methods ko Bootstrap buttons add karke live test kiya gaya taaki check ho sake ki styles browser par theek se render ho rahe hain ya nahi.

Next Tutorial Preview: Agle lecture mein Django ke andar **Tailwind CSS** integrate karna sikhaya jayega!